

ARJUN TYAGI

Urbana, IL · arjuntyagi25.github.io · arjun25@illinois.edu · 217-721-2414

SUMMARY

- Ph.D. student in Computer Architecture, specializing in cross-stack development and integration of processing-using-memory systems
- Extensive experience in memory systems, from devices to circuits, gained through multiple research internships

EDUCATION

University of Illinois Urbana-Champaign

Ph.D in Computer Science

Champaign, IL

Expected May 2029

Birla Institute of Technology & Science, Pilani

Bachelor of Engineering in Electronics and Instrumentation Engineering

Goa, India

Aug. 2019 - May 2023

PUBLICATIONS, PREPRINTS & TUTORIALS

Arjun Tyagi and Shubham Sahay, *Efficient Reinforcement Learning On Passive RRAM Crossbar Array*, Submitted to IEEE Transactions on Very Large Scale Integration Systems (TVLSI), preprint available upon request

Arjun Tyagi and Shahar Kvatinsky, *Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays*, 2024 IEEE International Symposium on Circuits and Systems (ISCAS), pp. 1-5, doi: 10.1109/ISCAS58744.2024.10558539

ARCANA Research Group and Sandia National Laboratories, *Tutorial on Simulation for Processing-Using-Memory Systems*, 2024 IEEE International Symposium on Computer Architecture (ISCA) Workshop

RESEARCH EXPERIENCE

ARCANA Research Group, PI: Prof. Saugata Ghose

Urbana, IL

Rapid, Accurate Chip-Scale Simulation of Digital Compute Using Emerging Memory Devices

Sept. 2023 - present

- Developed a fast simulation framework for digital in-memory computing, enabling chip-scale modeling with $> 10,000\times$ speedup over SPICE tools while maintaining high accuracy
- Introduced techniques such as memoization and dynamic time stepping to drastically reduce simulation time and memory usage while still being robust to device variability
- Enabled seamless integration of Verilog-A device models, supporting iterative co-design across device, circuit, and architecture levels
- Paper in preparation for submission to DAC 2026

ASIC² Research Group, PI: Prof. Shahar Kvatinsky

Haifa, Israel

Assessing the Performance of Stateful Logic in 1-Selector-1-RRAM Crossbar Arrays

Jan. - Jun. 2023

- Developed a compact Verilog-A model for 1S1R memory cells, aimed for large-scale simulations
- Modeled parasitic RC effects for intra-array interconnects to realistically evaluate the effect of array size on performance degradation
- Simulated passive (1R) and active (1S1R) RRAM arrays of varying sizes (4×4 to 512×512) for in-memory computing using MAGIC logic gates
- Demonstrated that 1S1R arrays offer up to $24.4\times$ lower power and $4.5\times$ higher readout margin than 1R arrays, enabling reliable scaling to large arrays

CtrlPIM: A Microcode-Based Controller for Memristive Memory Processing Unit

Jan. - Jun. 2023

- Co-designed a memory controller to support a PIM ISA for memristive memory processing unit (mMPU)
- Implemented the controller on an FPGA consisting of Zynq-7000 SoC

NeuroChaSe Group, PI: Prof. Shubham Sahay

Kanpur, India

Efficient Reinforcement Learning on Passive RRAM Crossbar Array

Jan. - Jul. 2022

- Performed an algorithm-hardware co-design and proposed a novel implementation of Monte Carlo reinforcement learning algorithm on passive RRAM crossbar array
- Demonstrated $> 100,000\times$ area reduction over prior digital and 1T1R-based implementations, while maintaining robustness to spatial/temporal variations and endurance failures in RRAM devices

AWARDS & SCHOLARSHIPS

- **First-Year Ph.D. Fellowship** *Aug. 2024 - May 2025*
University of Illinois Urbana-Champaign
- **Summer Undergraduate Research Internship** *May - Aug. 2022*
Arizona State University
- **Summer Research Fellowship Programme** *May - Aug. 2022*
Indian Institute of Technology Delhi

SKILLS & ABILITIES

- **Languages/Applications:** C, C++, Python, Verilog HDL, Verilog A, MATLAB
- **Graduate Courses:** Programming Languages, Compiler Construction, Computer System Organization, Parallel Computer Architectures, Emerging Memory & Storage Systems
- **Undergraduate Courses:** Microelectronic Circuits, Analog and Digital VLSI Design, Analog Electronics, Digital Design

TUTORING & VOLUNTEERING EXPERIENCE

- **Graduate Student Ambassador at UIUC CS** *Mar. 2025*
- **Undergraduate/Graduate Mentor, UIUC**
· Dylan Meeks, ECE '26 at University of Texas at Austin *Jun. - Aug. 2025*

TEACHING EXPERIENCE

- **ECE 313: Analog & Digital VLSI Design** **BITS Pilani**
Undergraduate Teaching Assistant *Aug. - Dec 2022*
- **ECE 241: Microprocessor & Interfacing** **BITS Pilani**
Undergraduate Teaching Assistant *Jan. - May 2022*
- **ECE 216: Digital Design** **BITS Pilani**
Undergraduate Teaching Assistant *Aug. - Dec 2021*